

Liar's Deck

User Manual

Bluff_Multiplayer — real-time multiplayer (desktop)

Prerequisites

- **Node.js** (LTS recommended) and **npm**
- Two terminal windows for local play (server + client)
- Modern desktop browser (Chrome / Edge / Firefox)

Optional: MongoDB Atlas or local MongoDB for room persistence after server restart
(`MONGODB_URI` in backend `.env`).

Install and run

1. Backend (port 3000)

```
cd src/backend  
npm install  
npm run dev
```

Or production build: `npm run build && npm start`

Default URL: `http://localhost:3000` (Socket.IO + Express).

Install and run (cont.)

2. Frontend (port 5173)

```
cd src/frontend  
npm install  
npm run dev
```

Open <http://localhost:5173> in the browser.

The app connects to `http://localhost:3000` (see `src/frontend/src/socket.ts`).
Change the URL there if the server runs elsewhere.

Lobby: create or join

1. Enter a **room code** (any non-empty string, max 32 characters).
2. **Create Room** — creates that room if it does not already exist.
3. **Join Room** — join an existing room (must exist, not full, game not started).

Errors (e.g. room full, wrong name) appear as **toasts** in the corner.

Room: name and ready

After joining:

1. Note the **Room Code**; use **Copy** to share with friends.
2. Enter **Your name** and click **Set Name** (max 20 characters).
3. When everyone is named, click **Ready Up**.

Start condition: At least **two** players in the room and **every** player must be ready. Then the server deals cards and revolvers and the game begins.

Leave returns you to the lobby and clears local session UI.

Playing the game

- **Declared rank:** A random table rank (King, Queen, or Ace) applies to the round; you play Liar cards onto the pile.
- **Your turn:** Select one or more cards in your hand, then **Play Cards**. Only the active player can play.
- **Call Liar:** When it is your turn and **Call Liar** is available, you may challenge the **previous** play. The pile is revealed; a roulette-style outcome may eliminate a player.
- Opponents see **hidden** hands (?); you only see your real cards after the server sends your hand via **getHand**.

Session and reconnect

- On create/join, the server issues a **session token** stored in `localStorage` as `sessionToken`.
- The Socket.IO client sends this token on connect; the server can **restore** your room and player when possible.
- Short disconnects are helped by Socket.IO **connection state recovery** (server configured with a recovery window).

If you see stale UI, refresh after ensuring the backend is running.

Optional: MongoDB persistence

Set in `src/backend/.env` (create if missing):

```
MONGODB_URI=mongodb+srv://...
```

When configured, room snapshots are saved to database `LiarsDeck`, collection `gameRooms`. If `MONGODB_URI` is unset, the game runs fully in memory.

Troubleshooting

Issue	What to try
Cannot connect	Start backend first; check firewall; URL in <code>socket.ts</code> matches server.
"Not your turn" / errors	Wait for your turn; toasts show server messages.
Game won't start	Need 2+ players and all ready.
CORS errors	Backend allows <code>http://localhost:5173</code> for Socket.IO.

Repository

https://github.com/joseph-ampfer/Bluff_Multiplayer

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